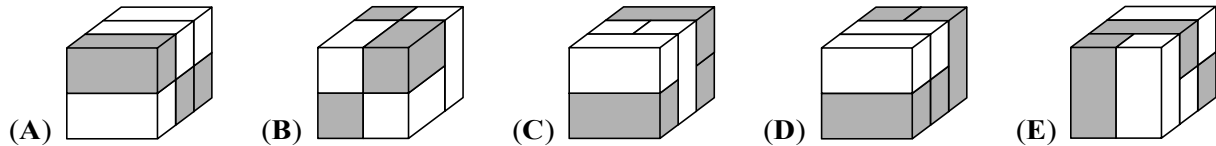
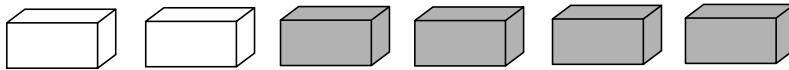


3 points

1. Which of the following solid shapes can be made with these 6 bricks ?

下列哪个立体图形可以用以下 6 块砖组成？



2. In how many places in the picture are two children holding each other with their left hands?

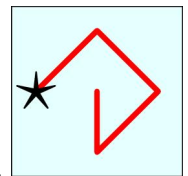
图中有几处是两个孩子互相用左手挽着对方？



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

1	2	3
4	5	6
7	8	9

3. In the square you can see the digits from 1 to 9. A number is created by starting



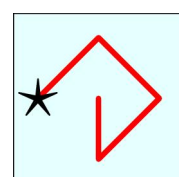
at the star, following the line and writing down the digits along the line while passing.

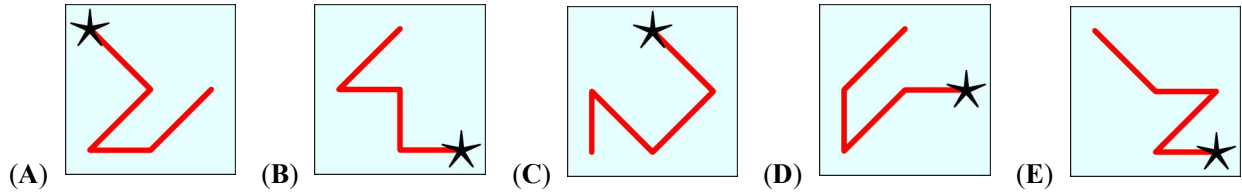
For example the line shown represents the number 42685.

Which of the following lines represents the largest number?

1	2	3
4	5	6
7	8	9

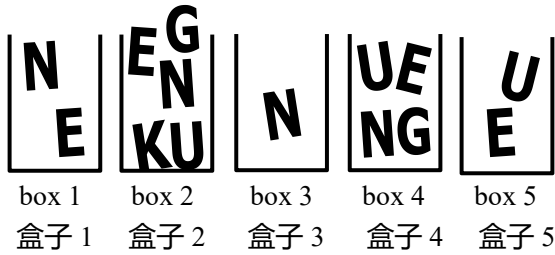
正方形中写有数字 1 至 9。从星号位置开始，沿着红线依次写下经过的数字，从而构成一个新的数。比如，右图红线表示的数为 42685。请问下列选项中哪条红线代表的数最大？





4. Sofie wants to write the word KENGU by using letters from the boxes. She can only take one letter from each box.

Sofie 想要用下面盒子中的字母写出单词 KENGU。她只能从每个盒子中取出一个字母。



What letter must Sofie take from box 4?

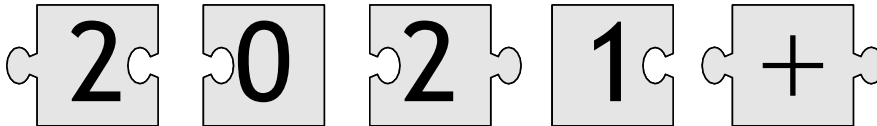
请问Sofie必须从盒子4中拿出哪个字母?

- (A) K (B) E (C) N (D) G (E) U

5. When the 5 pieces shown are fitted together correctly, the result is a rectangle with a calculation written on it.

What is the answer to this calculation?

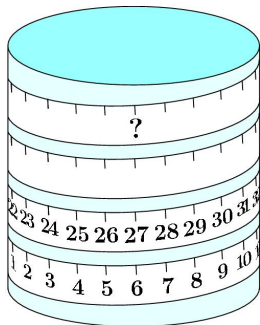
将下列 5 个拼图碎片正确拼接在一起后，会出现写有一道算式题的矩形。请问这个算式的答案是多少？



- (A) 22 (B) 32 (C) 41 (D) 122 (E) 203

6. A measuring tape is wound around a cylinder.

一把卷尺缠绕在一个圆柱体上。



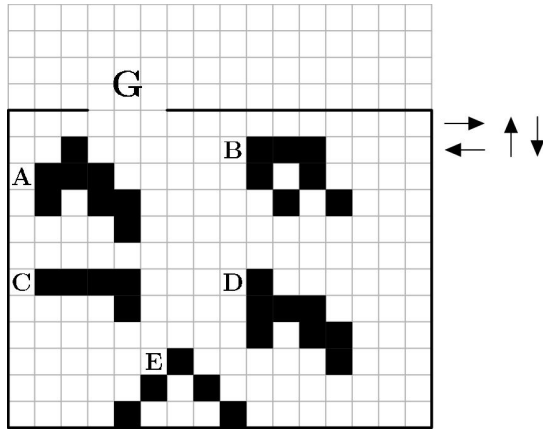
Which number should be at the place shown by the question mark?

请问图中问号处标记的应该是哪个数？

- (A) 53 (B) 60 (C) 69 (D) 77 (E) 81

7. The 5 figures on the grid can only move in the directions indicated by the black arrows. Which figure can leave through gate G?

下图网格中的 5 个图形只能按照黑色箭头所指方向进行移动。请问哪个图形可以从 G 门离开？



(A) A

(B) B

(C) C

(D) D

(E) E

8. Carin is going to paint the walls in her room green. The green paint is too dark so she mixes it with white paint. She tries different mixtures. Which of the following mixtures will give the darkest green colour?

Carin想要把她房间的墙面涂成绿色。考虑到绿色油漆很暗，她打算混合一些白色油漆。她尝试了不同的混合比例，请问下列哪种混合比例所呈现的绿色最暗？

(A) 1 part green + 3 parts white
1份绿色 + 3份白色

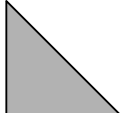
(B) 2 parts green + 6 parts white
2份绿色 + 6份白色

(C) 3 parts green + 9 parts white
3份绿色 + 9份白色

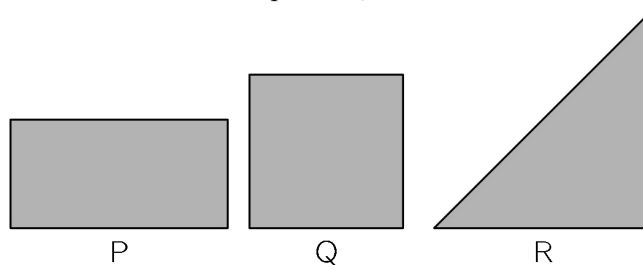
(D) 4 part green + 12 parts white
4份绿色 + 12份白色

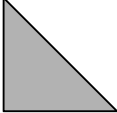
(E) They will all be equally dark
以上混合比例亮度相同

9. Mary had a piece of paper. She folded it exactly in half. Then she folded it exactly in half again.

She got this shape . Which of the shapes P, Q or R could have been the shape of her

original piece of paper?

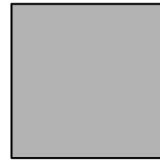


Mary将一张纸对折两次后，纸的形状变为 。请问在图形 P、Q 和 R 中，

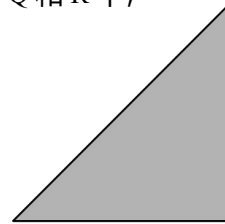
哪个图形可能是纸张对折前的形状？



P



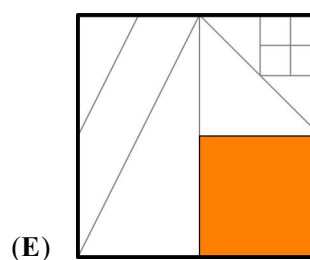
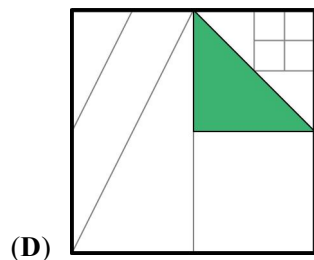
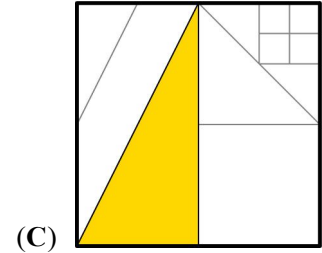
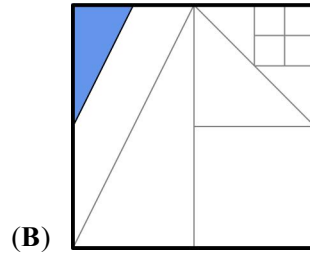
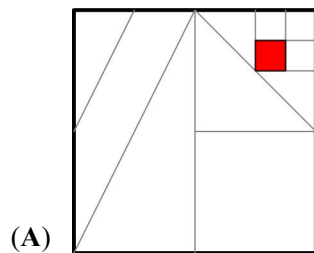
Q



R

- (A) only P 只有 P (B) only Q 只有 Q (C) only R 只有 R
(D) only P or Q 只有 P 和 Q (E) any of P, Q or R P、Q、R 均可

10. There is a square with line segments drawn inside it. The line segments are drawn either from the vertices or the midpoints of other line segments. We colored $\frac{1}{8}$ of the large square. Which one is our coloring?
正方形中画有线段，所有线段均为连接正方形顶点或其他线段的中点所得。我们将正方形的 $\frac{1}{8}$ 涂上了颜色，请问以下哪个是我们涂色的正方形？



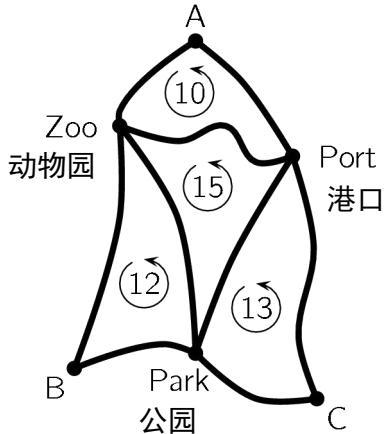
4 points

11. The number 5021972970 is written on a sheet of paper. Julian cuts the sheet twice so he gets three numbers.
What is the smallest sum he can get by adding these three numbers?
Julian 在纸上写下数 5021972970，然后把纸剪了两次，最终得到三个新的数。请问这三个数相加之和的最小值是多少？

- (A) 3244 (B) 3444 (C) 5172 (D) 5217 (E) 5444

- # 12. The map shows three bus stations at points A, B and C. A tour from station A to the Zoo and the Port and back to A is 10 km long. A tour from station B to the Park and the Zoo and back to B is 12 km long. A tour from station C to the Port and the Park and back to C is 13 km long. Also A tour from the Zoo to the Park and the Port and back to the Zoo is 15 km long. How long is the shortest tour from A to B to C and back to A?

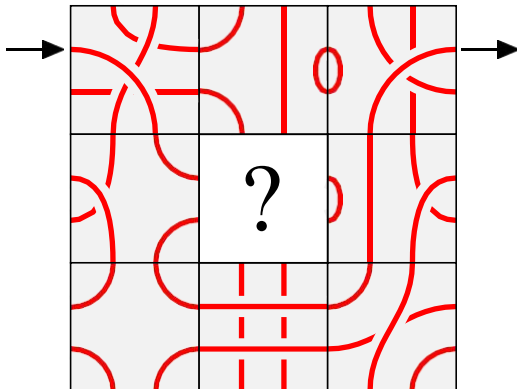
地图上显示了A、B、C三个公交站点。从A站到动物园再到港口，最后回到A站的路程是10 km。从B站再到公园和动物园，最后回到B站的路程是12 km。从C站到港口再到公园，最后回到C站的路程是13 km。从动物园到公园再到港口，最后回到动物园的路程是15 km。请问从A站到B站再到C站，最后回到A站的最短路程是多少？



- (A) 18 km (B) 20 km (C) 25 km (D) 35 km (E) 50 km

- # 13. Rosa wants to start at the arrow, follow the line, and get out at the other arrow.

Rosa 想要从左边箭头开始进入方格，沿着红色线条移动，并从右边箭头走出方格。



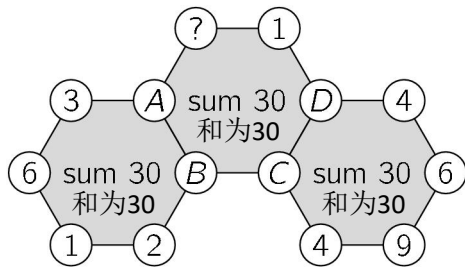
Which piece is it NOT possible to put in the middle to obtain that?

要符合上述走法，请问以下哪一块不能放在中间位置？

- (A) (B) (C) (D) (E)

- # 14. The diagram shows three hexagons with numbers at their vertices, but some numbers are invisible. The sum of the six numbers around each hexagon is 30. What is the number on the vertex marked with a question mark?

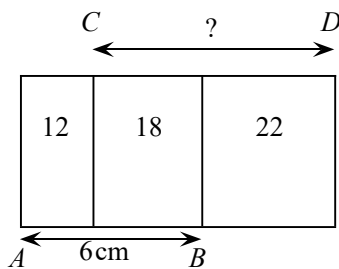
下图所示的三个六边形顶点上均标有数字，但有些数字看不见。已知每个六边形所有顶点上的数字之和为 30，请问标记有问号的顶点上的数字是多少？



- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

- # 15. Three rectangles of the same height are positioned as shown. The numbers within the rectangles indicate their areas in cm^2 . If $AB = 6 \text{ cm}$, how long is CD ?

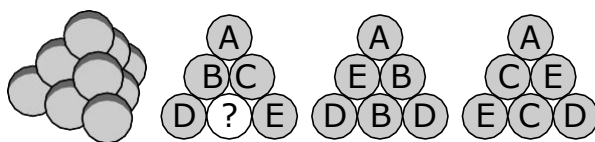
下图中三个矩形高度相同，每个矩形内部的数表示该矩形的面积 (cm^2)。如果 $AB = 6 \text{ cm}$ ，请问 CD 的长度是多少？



- (A) 7 cm (B) 7.5 cm (C) 8 cm (D) 8.2 cm (E) 8.5 cm

- # 16. A triangular pyramid is built with 10 identical balls, as shown. Each ball has one of the letters A, B, C, D and E on it. There are 2 balls marked with each letter. The picture shows three side views of the pyramid. What is the letter on the ball with the question mark?

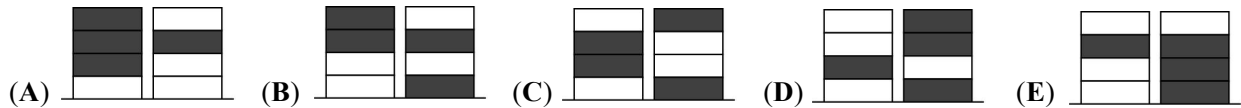
如图所示，10个相同的球搭建成一个三棱锥形状。每个球上标记有一个字母 A、B、C、D 或 E，每个字母均标记两个球，下图为三棱锥的三个侧面图。请问标记有问号的球上写有哪个字母？



- (A) A (B) B (C) C (D) D (E) E

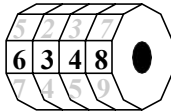
- # 17. Ronja had four white tokens and Wanja had four grey tokens. They played a game in which they took turns to place one of their tokens to create two piles. Ronja placed her first token first. Which pair of piles could they NOT create?

Ronja有四个白色代币，Wanja有四个灰色代币。他们玩了一个游戏，游戏中每人轮流放置一个代币，最后堆成两摞代币，Ronja先放置第一枚代币。请问下列哪个选项**不可能**是最后放置的结果？



- # 18. My little brother has a 4-digit bike lock with the digits 0 to 9 on each part of the lock as shown. He started on the correct combination and turned each part the same amount in the same direction and now the lock shows the combination 6348.

我弟弟的自行车锁是一个四位数滚轮密码锁，如下图所示，每个滚轮上均标有数字0到9。他从正确的解锁密码开始，将每个滚轮顺着同一方向转动相同的刻度后，密码锁上显示的数字组合是6348。



Which of the following CANNOT be the correct combination of my brother's lock?

请问以下哪一个**不可能**是弟弟自行车锁的密码？



- # 19. There were 20 apples and 20 pears in a box. Carl randomly took 20 pieces of fruit from the box and Luca took the rest. Which of the following statements is always true?

箱子中装有 20 个苹果和 20 个梨。Carl从箱子中随机拿走了20 个水果，Luca拿走了剩下的水果。

请问下列表述中，哪一个永远成立？

- (A) Carl got at least one pear. Carl 至少拿了一个梨
(B) Carl got as many apples as pears. Carl 拿的苹果和梨的数量一样多
(C) Carl got as many apples as Luca. Carl 和 Luca 拿的苹果的数量一样多
(D) Carl got as many pears as Luca got apples. Carl 拿的梨的数量和 Luca 拿的苹果的数量一样多
(E) Carl got as many pears as Luca. Carl 和 Luca 拿的梨的数量一样多

20. There is a single train track between points X and Y .

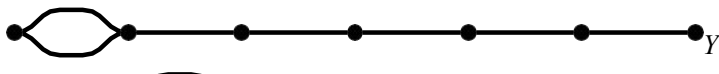
在 X 站和 Y 站之间只有一条铁轨。



A train company wants one train to leave from X and one train to leave from Y at the same time daily. Moving with constant speed it takes 180 minutes for a train to make a trip from X to Y and 60 minutes from Y to X .

They want to build a double track  to avoid a crash. Where should the double track be?

铁路公司打算每天同一时间从 X 站发出一列火车，从 Y 站发出另一辆火车。两辆火车匀速行驶，已知从 X 站到 Y 站需要 180 分钟，从 Y 站到 X 站需要 60 分钟。公司打算再修一条复线铁路  以避免撞车。请问这条复线铁路应该修在哪里？

- (A) 
- (B) 
- (C) 
- (D) 
- (E) 

5 points

21. Ann, Bob, Carina, Dan and Ed are sitting at a round table. Ann is not next to Bob, Dan is next to Ed and Bob is not next to Dan. Which two people are sitting next to Carina?

Ann, Bob, Carina, Dan 和 Ed 围坐在一张圆桌上。Ann 没有坐在 Bob 旁边，Dan 坐在 Ed 旁边，Bob 没有坐在 Dan 旁边。请问哪两个人坐在 Carina 旁边？

- (A) Ann and Bob Ann 和 Bob (B) Bob and Dan Bob 和 Dan (C) Dan and Ed Dan 和 Ed
(D) Ed and Ann Ed 和 Ann (E) It is not possible to be certain 无法确定

22. Maurice asked the canteen chef for the recipe for his pancakes.

Maurice向食堂厨师请教了摊煎饼的方法。

Ingredients for 100 pancakes (摊100个煎饼所需配料)
25 eggs (25个鸡蛋) 4 l milk (4升牛奶)
5 kg flour (5千克面粉) 1 kg butter (1千克黄油)

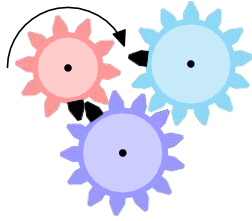
Maurice has 6 eggs, 400g flour, 0,5 liters of milk and 200g butter. What is the largest number of pancakes he can make using this recipe?

Maurice现在有 6 个鸡蛋，400 克面粉，0.5 升牛奶和 200 克黄油。按照上面的配方摊煎饼，他最多能做出多少个煎饼？

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 15

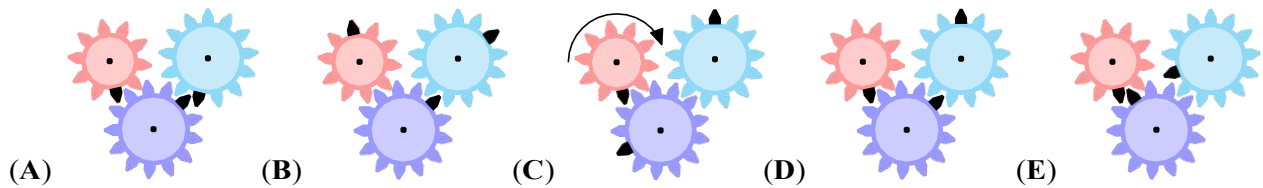
23. The picture shows three gears with a black gear tooth on each.

下图中有三个齿轮，每个齿轮上有一个黑色的轮齿。



Which picture shows the correct position of the black teeth after the small gear has turned a full turn clockwise?

下列哪张图片显示的是小齿轮按顺时针方向旋转一周后，所有黑色齿轮的正确位置？

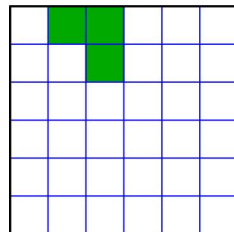


24. An apple and an orange weigh as much as a pear and a peach. An apple and a pear weigh less than an orange and a peach, and a pear and an orange weigh less than an apple and a peach. Which of the pieces of fruit is the heaviest?

一个苹果加一个橘子的重量等于一个梨加一个桃子的重量。一个苹果加一个梨的重量小于一个橘子加一个桃子的重量。一个梨加一个橘子的重量小于一个苹果加一个桃子的重量。请问哪个水果最重？

- (A) apple 苹果 (B) orange 橘子 (C) peach 桃子 (D) pear 梨
(E) impossible to determine 无法确定

25. What is the smallest number of shaded squares that can be added to the diagram to create a



design with four axes of symmetry?

请问至少再给多少个小正方形涂色，才能使得以上整个图案有四条对称轴？

- (A) 1 (B) 9 (C) 12 (D) 13 (E) 21

- # 26. Three pirates were asked how many coins and how many diamonds their friend Graybeard had. Each of the three told the truth to one question but told a lie to the other. Their answers are written on the piece of paper pictured. What is the total number of coins and diamonds that Graybeard has?

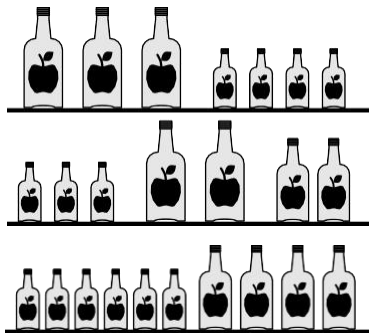
三个海盗被问到他们的朋友灰胡子有多少硬币和多少钻石时，他们都只对其中一个问题说了实话，对另一个问题撒了谎。他们的回答记录在下面纸片上。请问灰胡子拥有的硬币和钻石总数是多少？

- (1) He has 8 coins and 6 diamonds. 他有8枚硬币和6颗钻石。
(2) He has 7 coins and 4 diamonds. 他有7枚硬币和4颗钻石。
(3) He has 7 coins and 7 diamonds. 他有7枚硬币和7颗钻石。

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

- # 27. Each shelf holds a total of 64 deciliters of apple juice. The bottles have three different sizes: large, medium and small. How many deciliters of apple juice does a medium bottle contain?

下图每层货架上均放置了 64 分升的苹果汁，装苹果汁的瓶子有大号、中号、小号三种不同规格。请问一个中号瓶子可以装多少分升苹果汁？



- (A) 3 (B) 6 (C) 8 (D) 10 (E) 14

- # 28. A large cube has side-length 7cm. On each of its 6 faces, the two diagonals are drawn in red. The large cube is then cut into small cubes with side-length 1cm. How many small cubes will have at least one red line drawn on it?

一个大立方体的棱长为 7 厘米。将这个立方体每个面上的两条对角线都涂红，然后把大立方体切成棱长为 1 厘米的小立方体。请问有多少个小立方体至少画有一条红线？

- (A) 54 (B) 62 (C) 70 (D) 78 (E) 86

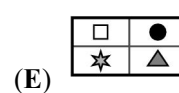
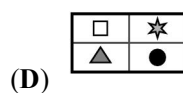
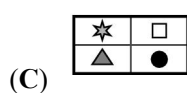
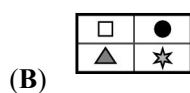
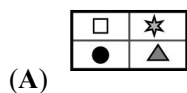
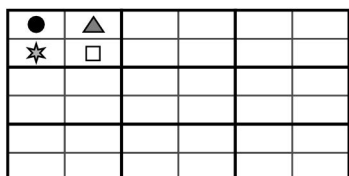
- # 29. In a group of 10 elves and trolls, each were given a token with a different number from 1 to 10 written upon it. They were each asked what number was on their token and all answered with a number from 1 to 10. The sum of the answers was 36. Each troll told a lie and each elf told the truth. What is the smallest number of trolls there could be in the group?"

小组中有精灵和巨魔共 10 只。每个精灵和巨魔都得到了一个代币，代币上面写着从 1 到 10 中的一个数字（不重复）。当他们被问到代币上的数时，每个巨魔都说了谎，每个精灵都说了实话。已知他们回答的数都在 1 到 10 的范围内，而且回答的数字之和是 36。请问小组中最少有多少只巨魔？

- (A) 1 (B) 3 (C) 4 (D) 5 (E) 7

- # 30. There are rectangular cards divided into four equal cells with different shapes $\square$, $\star$, $\bullet$, $\blacktriangle$ drawn in each cell. Cards can be placed side by side only if the same shapes appear in adjacent cells on their common side. Nine cards are used to form a rectangle as shown in the figure. Which of the following cards was definitely NOT used to form this rectangle?

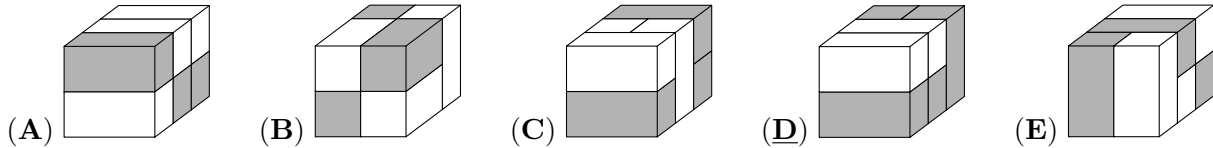
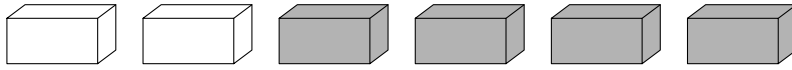
将矩形卡片等分为四个单元格，每个单元格中画有 $\square$ 、 $\star$ 、 $\bullet$ 、 $\blacktriangle$ 四种不同形状。规定只有两张卡片公共边上的相邻单元格中图形相同时，这两张卡片才可以并排放置。如下图所示，用九张卡片组成一个大矩形。请问以下哪张卡片**不能**用来组成这个大矩形？



PROBLEMS WITH SOLUTIONS

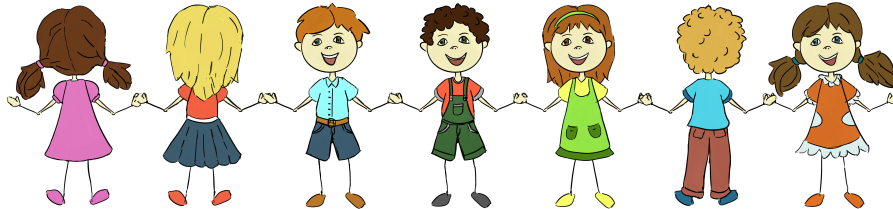
3 points

1. Which of the following solid shapes can be made with these 6 bricks ?



SOLUTION:

2. In how many places in the picture are two children holding each other with their left hands?



- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

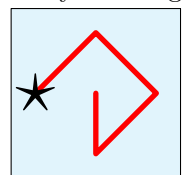
SOLUTION:

1	2	3
4	5	6
7	8	9

3. In the square you can see the digits from 1 to 9.

7	8	9
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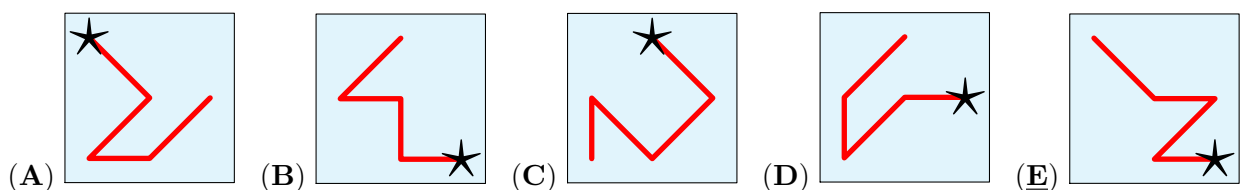
 A number is created by starting



at the star, following the line and writing down the digits along the line while passing.

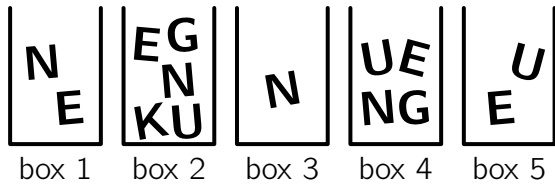
For example the line shown represents the number 42685.

Which of the following lines represents the largest number?



SOLUTION:

4. Sofie wants to write the word KENGU by using letters from the boxes. She can only take one letter from each box.

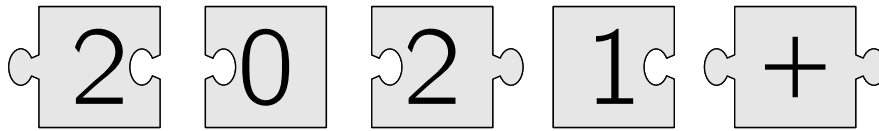


What letter must Sofie take from box 4?

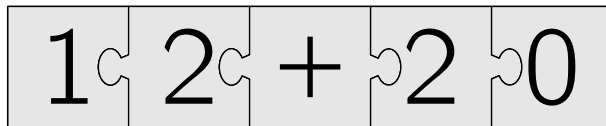
- (A) K (B) E (C) N (D) G (E) U

SOLUTION:

5. When the 5 pieces shown are fitted together correctly, the result is a rectangle with a calculation written on it. What is the answer to this calculation?

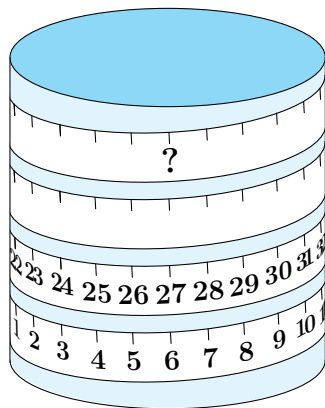


- (A) 22 (B) 32 (C) 41 (D) 122 (E) 203



SOLUTION:

6. A measuring tape is wound around a cylinder.

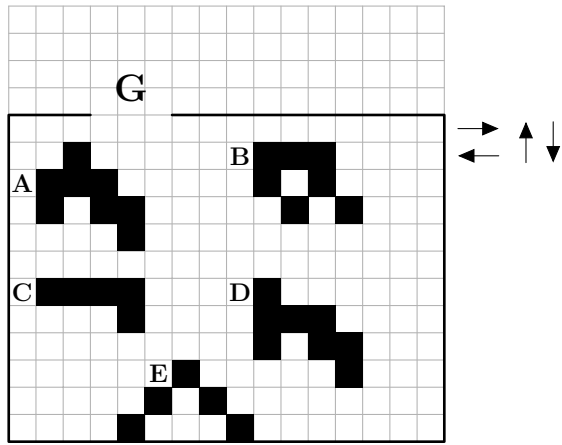


Which number should be at the place shown by the question mark?

- (A) 53 (B) 60 (C) 69 (D) 77 (E) 81

SOLUTION:

7. The 5 figures on the grid can only move in the directions indicated by the black arrows. Which figure can leave through gate G?



(A) A

(B) B

(C) C

(D) D

(E) E

SOLUTION:

8. Carin is going to paint the walls in her room green. The green paint is too dark so she mixes it with white paint. She tries different mixtures. Which of the following mixtures will give the darkest green colour?

(A) 1 part green + 3 parts white

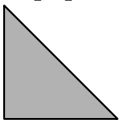
(B) 2 parts green + 6 parts white

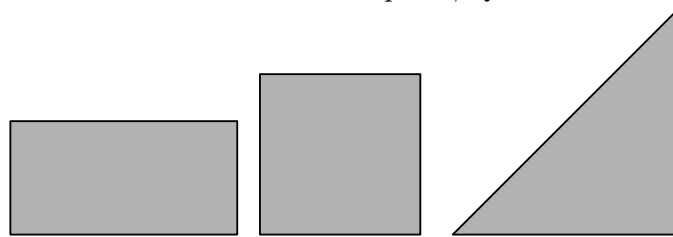
(C) 3 parts green + 9 parts white

(D) 4 part green + 12 parts white

(E) They will all be equally dark

SOLUTION: They are all the same proportion

9. Mary had a piece of paper. She folded it exactly in half. Then she folded it exactly in half again. She got this shape . Which of the shapes P, Q or R could have been the shape of her



original piece of paper?

(A) only P

(B) only Q

(C) only R

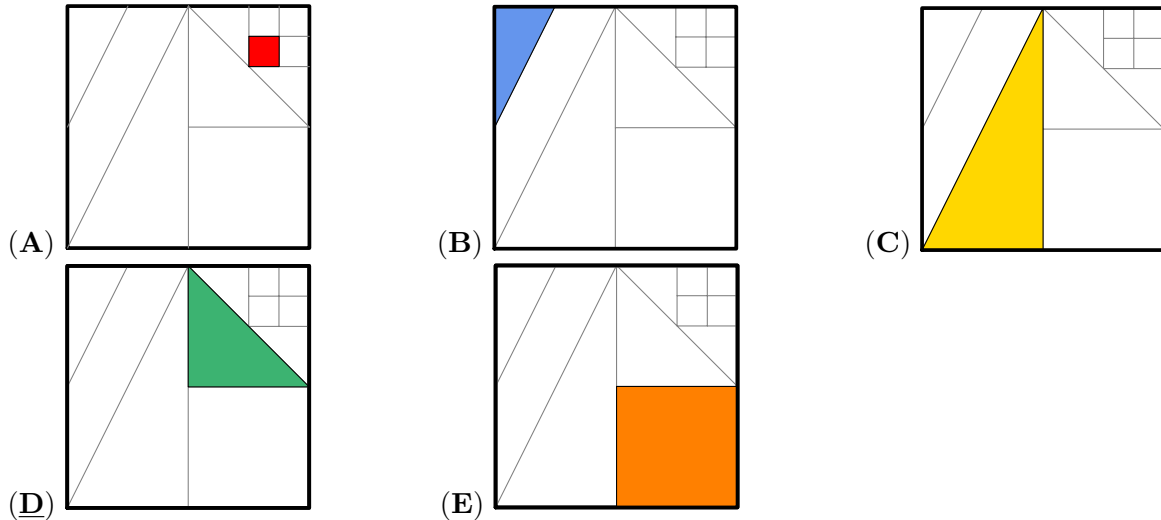
(D) only P or Q

(E) any of P, Q or R

SOLUTION:

10. There is a square with line segments drawn inside it. The line segments are drawn either from the vertices or the midpoints of other line segments. We colored $\frac{1}{8}$ of the large square. Which one is

our coloring?



SOLUTION:

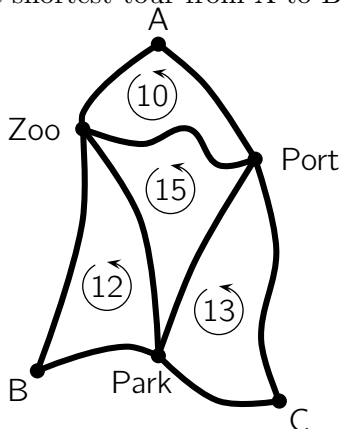
4 points

11. The number 5021972970 is written on a sheet of paper. Julian cuts the sheet twice so he gets three numbers. What is the smallest sum he can get by adding these three numbers?

- (A) 3244 (B) 3444 (C) 5172 (D) 5217 (E) 5444

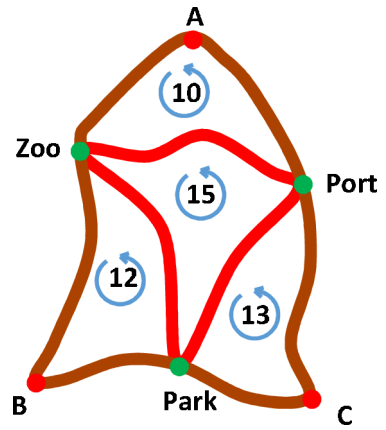
SOLUTION: The four figure number we utilise must be as small as possible. Answer is $502 + 1972 + 970 = 3444$.

12. The map shows three bus stations at points A, B and C. A tour from station A to the Zoo and the Port and back to A is 10 km long. A tour from station B to the Park and the Zoo and back to B is 12 km long. A tour from station C to the Port and the Park and back to C is 13 km long. Also A tour from the Zoo to the Park and the Port and back to the Zoo is 15 km long. How long is the shortest tour from A to B to C and back to A?



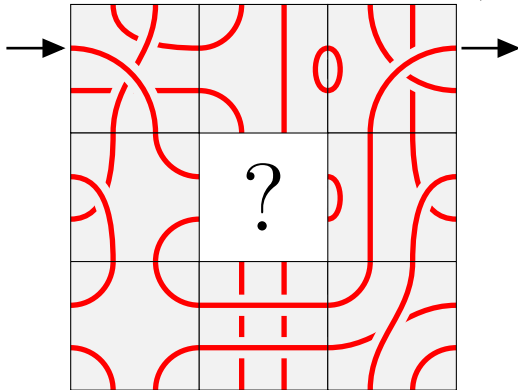
- (A) 18 km (B) 20 km (C) 25 km (D) 35 km (E) 50 km

SOLUTION: If we add the lengths of the described round trips from A, B and C we get the outer path (brown line) plus the inner path (red line). So if from this we subtract the inner path, we get the

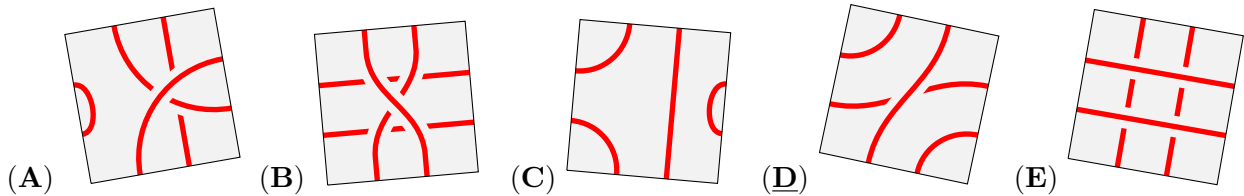


length of the required outer path, namely $10+12+13-15=20$ km.

13. Rosa wants to start at the arrow, follow the line, and get out at the other arrow.

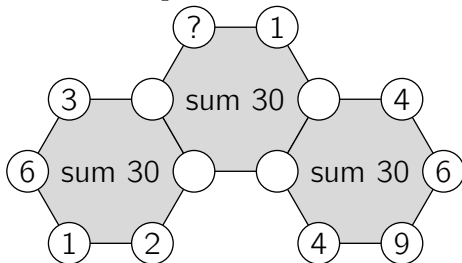


Which piece is it NOT possible to put in the middle to obtain that?



SOLUTION:

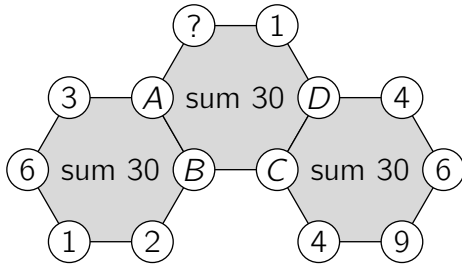
14. The diagram shows three hexagons with numbers at their vertices, but some numbers are invisible. The sum of the six numbers around each hexagon is 30. What is the number on the vertex marked with a question mark?



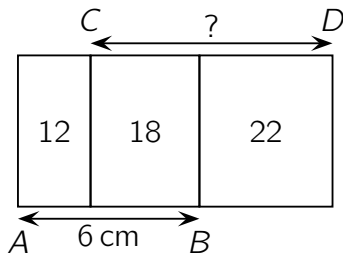
- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

SOLUTION: The sum of the visible numbers on the left hexagon is $3 + 6 + 1 + 2 = 12$. As the sum of all numbers on the hexagon is 30, it means that the sum of the numbers at vertices A and B is

$30 - 12 = 18$. Similarly on the right hexagon the visible numbers add up to 23 so the sum of the numbers at vertices C and D is $30 - 23 = 7$. Now look at the central hexagon. The sum of the numbers at A , B , C , D and the 1 at the top right vertex is $18 + 7 + 1 = 26$. So what remains is $30 - 26 = 4$.



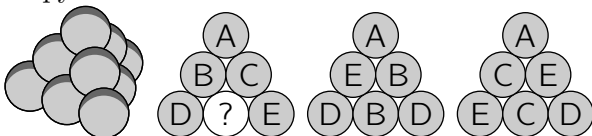
15. Three rectangles of the same height are positioned as shown. The numbers within the rectangles indicate their areas in cm^2 . If $AB = 6$ cm, how long is CD ?



- (A) 7 cm (B) 7.5 cm (C) 8 cm (D) 8.2 cm (E) 8.5 cm

SOLUTION: From the combined area of the first two rectangles we see that their height $h = (12 + 18)/6 = 5$ cm. So $CD = (18 + 22)/5 = 8$ cm.

16. A triangular pyramid is built with 10 identical balls, as shown. Each ball has one of the letters A, B, C, D and E on it. There are 2 balls marked with each letter. The picture shows three side views of the pyramid. What is the letter on the ball with the question mark?



- (A) A (B) B (C) C (D) D (E) E

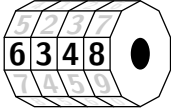
SOLUTION: The ball on the top of the pyramid is shown in all three side views. It has the letter A on it. The second ball with A on it has to appear on at least one of the side views. Since this is not the case, it must be the one with the question mark.

17. Ronja had four white tokens and Wanja had four grey tokens. They played a game in which they took turns to place one of their tokens to create two piles. Ronja placed her first token first. Which pair of piles could they not create?

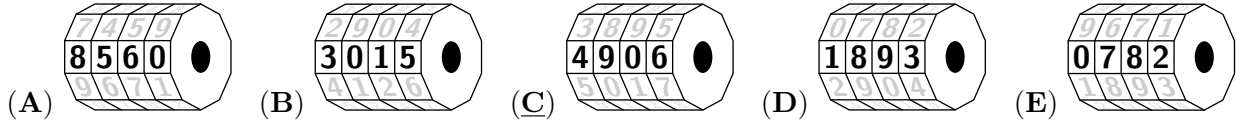


SOLUTION:

18. My little brother has a 4-digit bike lock with the digits 0 to 9 on each part of the lock as shown. He started on the correct combination and turned each part the same amount in the same direction and now the lock shows the combination 6348.



Which of the following CANNOT be the correct combination of my brother's lock?



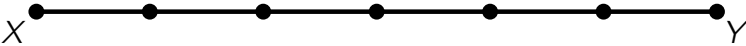
SOLUTION:

19. There were 20 apples and 20 pears in a box. Carl randomly took 20 pieces of fruit from the box and Luca took the rest. Which of the following statements is always true?

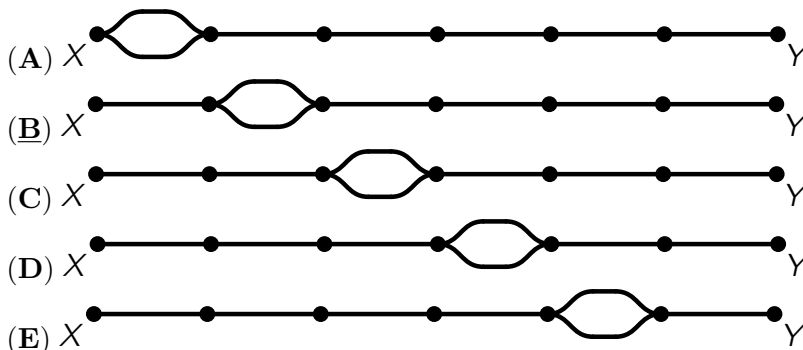
- (A) Carl got at least one pear. (B) Carl got as many apples as pears.
 (C) Carl got as many apples as Luca. (D) Carl got as many pears as Luca got apples.
 (E) Carl got as many pears as Luca.

SOLUTION: The number of Carl's apples plus the number of Luca's apples equals 20. The number of Carl's apples plus the number of Carl's pears also equals 20. Hence, Carl got as many pears as Luca got apples. Other answers are not always true.

20. There is a single train track between points X and Y .



A train company wants one train to leave from X and one train to leave from Y at the same time daily. Moving with constant speed it takes 180 minutes for a train to make a trip from X to Y and 60 minutes from Y to X . They want to build a double track  to avoid a crash. Where should the double track be?



SOLUTION:

5 points

21. Ann, Bob, Carina, Dan and Ed are sitting at a round table. Ann is not next to Bob, Dan is next to Ed and Bob is not next to Dan. Which two people are sitting next to Carina?

- (A) Ann and Bob (B) Bob and Dan (C) Dan and Ed (D) Ed and Ann
(E) It is not possible to be certain

SOLUTION:

22. Maurice asked the canteen chef for the recipe for his pancakes.

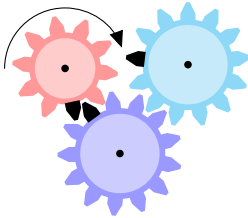
Ingredients for 100 pancakes			
25 eggs	4 l milk		
5 kg flour	1 kg butter		

Maurice has 6 eggs, 400g flour, 0,5 liters of milk and 200g butter. What is the largest number of pancakes he can make using this recipe?

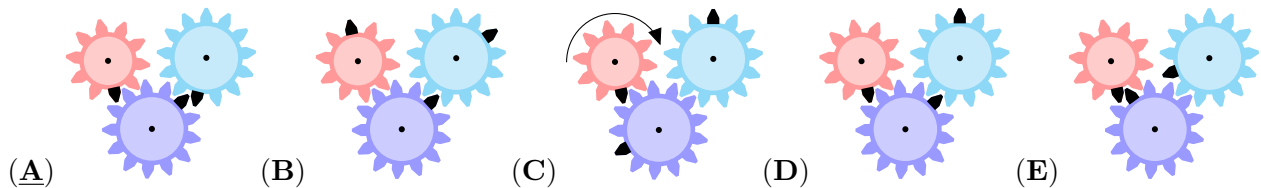
- (A) 6 (B) 8 (C) 10 (D) 12 (E) 15

SOLUTION:

23. The picture shows three gears with a black gear tooth on each.



Which picture shows the correct position of the black teeth after the small gear has turned a full turn clockwise?



SOLUTION: The black tooth of the first gear moves 10 positions clockwise. So the black tooth on each of the other gears also move 10 positions, but anticlockwise for the middle gear and clockwise for the top one. Counting 10 teeth away, in the appropriate direction, we easily locate the final positions.

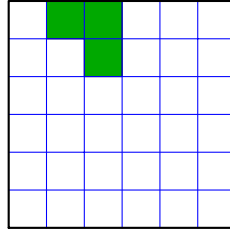
24. An apple and an orange weigh as much as a pear and a peach. An apple and a pear weigh less than an orange and a peach, and a pear and an orange weigh less than an apple and a peach. Which of the pieces of fruit is the heaviest?

- (A) apple (B) orange (C) peach (D) pear
(E) impossible to determine

SOLUTION: We have $(Ap + Or = Pr + Pch)$ and also $(Ap + Pe < Or + Pch)$. Adding we get, after the simplification, $(Ap < Pch)$. Also, subtracting the first equation from the inequality we get, after the

simplification, $(Pe < Or)$. Similarly from the first equality and the condition $(Pe + Or < Ap + Pch)$ we get $(Or < Pch)$ and $(Pe < A)$. Summing up, these conditions are written $(Pe < A < Pch)$ and $(Pe < Or < Pch)$. So the heaviest is Pch .

25. What is the smallest number of shaded squares that can be added to the diagram to create a

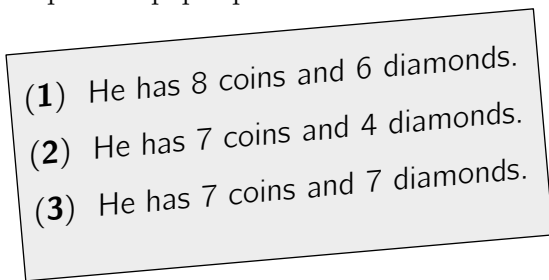


design with four axes of symmetry?

- (A) 1 (B) 9 (C) 12 (D) 13 (E) 21

SOLUTION:

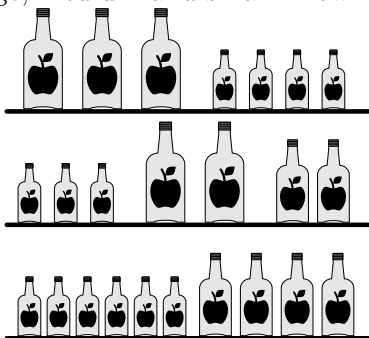
26. Three pirates were asked how many coins and how many diamonds their friend Graybeard had. Each of the three told the truth to one question but told a lie to the other. Their answers are written on the piece of paper pictured. What is the total number of coins and diamonds that Graybeard has?



- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

SOLUTION: The first statement of pirate 2 and pirate 3 were false (7 coins) then their second statements would both be true. But this cannot happen as they are different. So the friend had 7 coins. From pirate 1 we now see that the friend had 6 diamonds.

27. Each shelf holds a total of 64 deciliters of apple juice. The bottles have three different sizes: large, medium and small. How many deciliters of apple juice does a medium bottle contain?



- (A) 3 (B) 6 (C) 8 (D) 10 (E) 14

SOLUTION: From Solving a standard linear system we will find $L = 16$, $M = 10$, $S = 4$.

28. A large cube has side-length 7cm. On each of its 6 faces, the two diagonals are drawn in red. The large cube is then cut into small cubes with side-length 1cm. How many small cubes will have at least one red line drawn on it?

- (A) 54 (B) 62 (C) 70 (D) 78 (E) 86

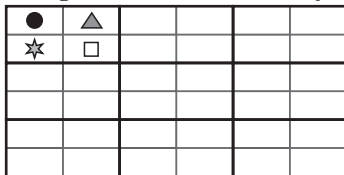
SOLUTION: From Michael: Each diagonal has 7 cubes with a line so each face has $7 + 7 - 1 = 13$ (the "-1" is for double counting the central square on the face). So all 6 faces have $13 \cdot 6$ minus the twice corner cubes that have been treble counted. So total is $13 \cdot 6 - 2 \cdot 8 = 62$. Alternatively we can count 9 internal cubes (i.e. not on the edge) on each face, plus the 8 corner ones. Total $9 \cdot 6 + 12 = 62$.

29. In a group of 10 elves and trolls, each were given a token with a different number from 1 to 10 written upon it. They were each asked what number was on their token and all answered with a number from 1 to 10. The sum of the answers was 36. Each troll told a lie and each elf told the truth. What is the smallest number of trolls there could be in the group?"

- (A) 1 (B) 3 (C) 4 (D) 5 (E) 7

SOLUTION:

30. There are rectangular cards divided into four equal cells with different shapes $\square$, $\star$, $\bullet$, $\blacktriangle$ drawn in each cell. Cards can be placed side by side only if the same shapes appear in adjacent cells on their common side. Nine cards are used to form a rectangle as shown in the figure. Which of the following cards was definitely NOT used to form this rectangle?



- (A)

$\square$	$\star$
$\bullet$	$\blacktriangle$

 (B)

$\square$	$\bullet$
$\blacktriangle$	$\star$

 (C)

$\star$	$\square$
$\blacktriangle$	$\bullet$

 (D)

$\square$	$\star$
$\blacktriangle$	$\bullet$

 (E)

$\square$	$\bullet$
$\star$	$\blacktriangle$

SOLUTION: